

Rethinking Kullback-Leibler Divergence in Knowledge Distillation for Large Language Models

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Date: 2024.09.17

Introduction

Prior assertion: reverse Kullback-Leibler (RKL) divergence is **mode-seeking** and thus preferable over the **mean-seeking** forward Kullback-Leibler (FKL) divergence.

This study: **neither** mode-seeking nor mean-seeking properties manifest in KD for LLMs.

- RKL and FKL **share the same optimization objective** and both converge after a sufficient number of epochs.
- RKL focuses on the **tail** part of the distributions, while FKL focuses on the **head** part.
- Propose Adaptive Kullback-Leiber (AKL) divergence, which adaptively allocates weights to combine FKL and RKL.

Background

Large model size (LLMs) brings expensive computing costs and high latency, it is crucial to compress the model size while keeping the performance as much as possible

Knowledge Distillation (KD) trains a **compact** student model by mimicking the behaviors of the (large) teacher model.

KL Divergence is used as a loss function in KD.

KL Divergence

Definition

$$D_{KL}(P||Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(X)}{Q(X)} \right]$$

Measure the distance between two distribution.

Continuous case: $\int p(x) \log \frac{p(x)}{q(x)} dx$

Discrete case: $\sum_x p(x) \log \frac{p(x)}{q(x)}$

Property

- Not symmetric: $D_{KL}(P||Q) \neq D_{KL}(Q||P)$, as a result, it is not a distance metric.
- $D_{KL}(P||Q) \in [0, \infty]$
- $D_{KL}(P||Q) = 0 \Leftrightarrow P = Q$

KL Divergence

$$D_{KL}(P||Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(X)}{Q(X)} \right]$$

Forward KL (FKL) divergence

$$FKL(p, q_{\theta}) = D_{KL}(p||q_{\theta}) = \sum_z p(z) \log \frac{p(z)}{q_{\theta}(z)}$$

Reverse KL (RKL) divergence

$$RKL(p, q_{\theta}) = D_{KL}(q_{\theta}||p) = \sum_z q_{\theta}(z) \log \frac{q_{\theta}(z)}{p(z)}$$

Previous claim

FKL causes the mean-seeking behavior where q_{θ} tends to cover the main mass of p simultaneously

RKL causes the mode-seeking behavior where q_{θ} tends to select one mode

RKL is more suitable than FKL to align the logits from LLMs since RKL avoids learning too many long-tail variants

Mean- and Mode-Seeking

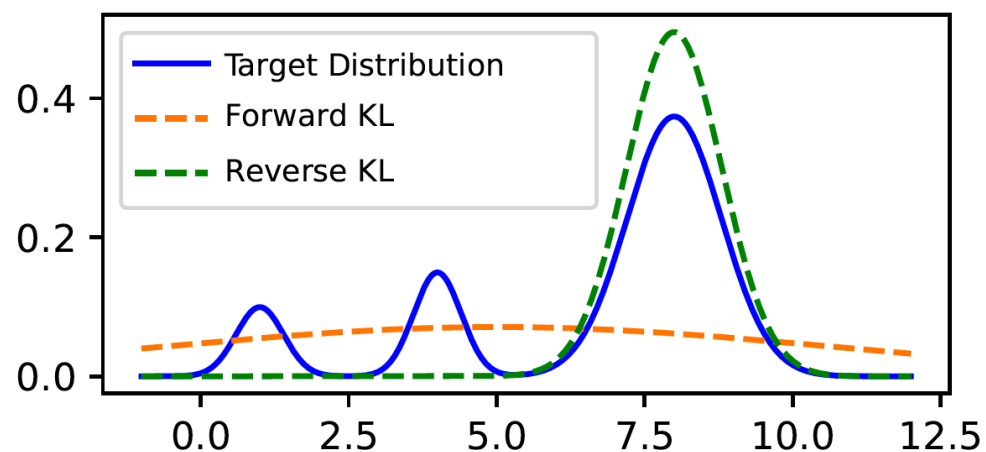


Figure 1: The toy example in Gu et al. (2023), where they fit a Gaussian mixture (distribution of teacher) with a Gaussian distribution (distribution of student) using FKL and RKL.

$$D_{KL}(P||Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(X)}{Q(X)} \right] = \int p(x) \log \frac{p(x)}{q(x)} dx$$

- Mean-Seeking Behaviour (FKL)

Wherever $P(\cdot)$ has high probability, $Q(\cdot)$ must also have high probability.

- Mode-Seeking Behaviour (RKL)

Wherever $Q(\cdot)$ has high probability, $P(\cdot)$ must also have high probability.

Mean- and Mode-Seeking

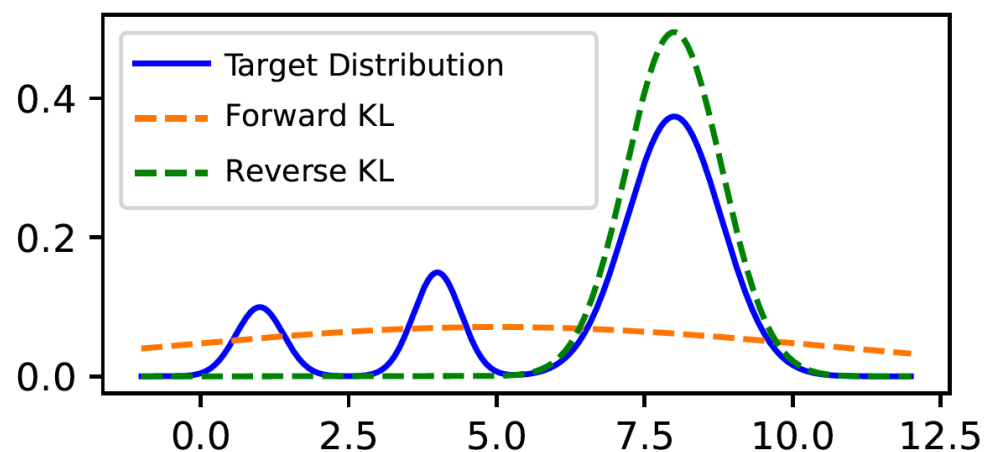


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Two assumptions:

- The student distribution q is unimodal such as the Gaussian distribution
- Teacher distribution p and student distribution q are continuous

KL in the KD for LLMs

Two assumptions do not hold

- q is not guaranteed to be unimodal
- p and q are discrete

RKL and FKL both converge after a sufficient number of epochs.

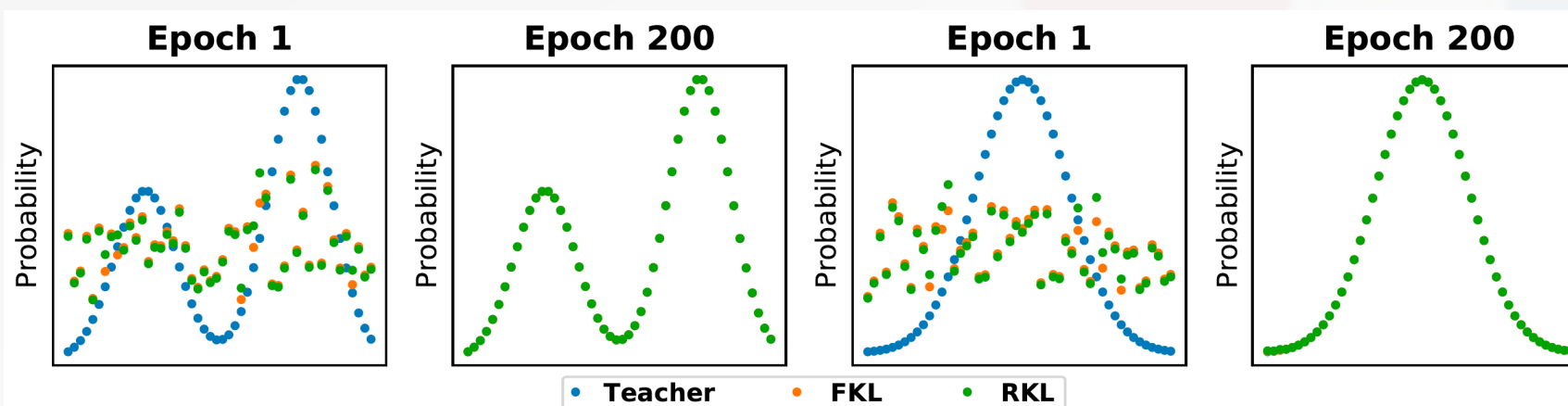


Figure 2: The convergence of FKL and RKL on toy data under epoch 1 and epoch 200. The initial distribution q is the same for FKL and RKL. After 200 epochs, both FKL and RKL can converge to the target distribution well regardless of the shape of p .

KL in the KD for LLMs

RKL and FKL share the same optimization objective

The p and q_θ is calculated by the softmax function

$$p(Y_j | \mathbf{y}_{<t}) = \frac{\exp(z_j^p)}{\sum_{k=1}^V \exp(z_k^p)}; \quad (3)$$

$$q_\theta(Y_j | \mathbf{y}_{<t}) = \frac{\exp(z_j^q)}{\sum_{k=1}^V \exp(z_k^q)}. \quad (4)$$

Gradient

$$\frac{\partial FKL(p, q_\theta)}{\partial z_j^q} = q_\theta(Y_j | \mathbf{y}_{<t}) - p(Y_j | \mathbf{y}_{<t}) \quad (5)$$

$$\frac{\partial RKL(p, q_\theta)}{\partial z_j^q} = q_\theta(Y_j | \mathbf{y}_{<t}) (\log(\frac{q_\theta(Y_j | \mathbf{y}_{<t})}{p(Y_j | \mathbf{y}_{<t})}) - RKL(p, q_\theta)). \quad (6)$$

Considering the converge condition:

$$\frac{\partial F(R)KL(p, q_\theta)}{\partial z_j^q} = 0 \quad \forall j \in \{1, 2, 3, \dots, V\}, \quad (7)$$

For both FKL and RKL, the sufficient and necessary condition for converge is

$$q_\theta(Y_j | \mathbf{y}_{<t}) = p(Y_j | \mathbf{y}_{<t}) \quad \forall j \in \{1, 2, 3, \dots, V\}. \quad (8)$$

FKL: obvious

RKL: show sufficiency and necessity

Sufficiency: obvious

Necessity:
$$q_\theta(Y_i | \mathbf{y}_{<t}) \frac{\partial RKL(p, q_\theta)}{\partial z_j^q} - q_\theta(Y_j | \mathbf{y}_{<t}) \frac{\partial RKL(p, q_\theta)}{\partial z_i^q} = q_\theta(Y_i | \mathbf{y}_{<t}) * 0 - q_\theta(Y_j | \mathbf{y}_{<t}) * 0 = 0. \quad (22)$$

KL in the KD for LLMs

The difference between FKL and RKL in the KD for LLMs: the **optimization process**

$$\text{FKL: } \sum_z p(z) \log \frac{p(z)}{q_\theta(z)}$$

- Larger $p(z)$ means a larger weight in total loss and also more likely to generate a larger $\log \frac{p(z)}{q_\theta(z)}$
- Fitting the area with larger $p(z)$ (a.k.a. the **head** part) is the priority of FKL

$$\text{RKL: } \sum_z q_\theta(z) \log \frac{q_\theta(z)}{p(z)}$$

- When $p(z)$ is smaller, $\frac{q_\theta(z)}{p(z)}$ is easier to be $+\infty$, leading to a larger loss
- fitting the area with smaller $p(z)$ (a.k.a. the **tail** part) is the priority of RKL

LLMs are seldom trained for such an extensive number of epoch, RKL fits better at the tail part and FKL fits better at the head part **at beginning epochs**.

Validation Example

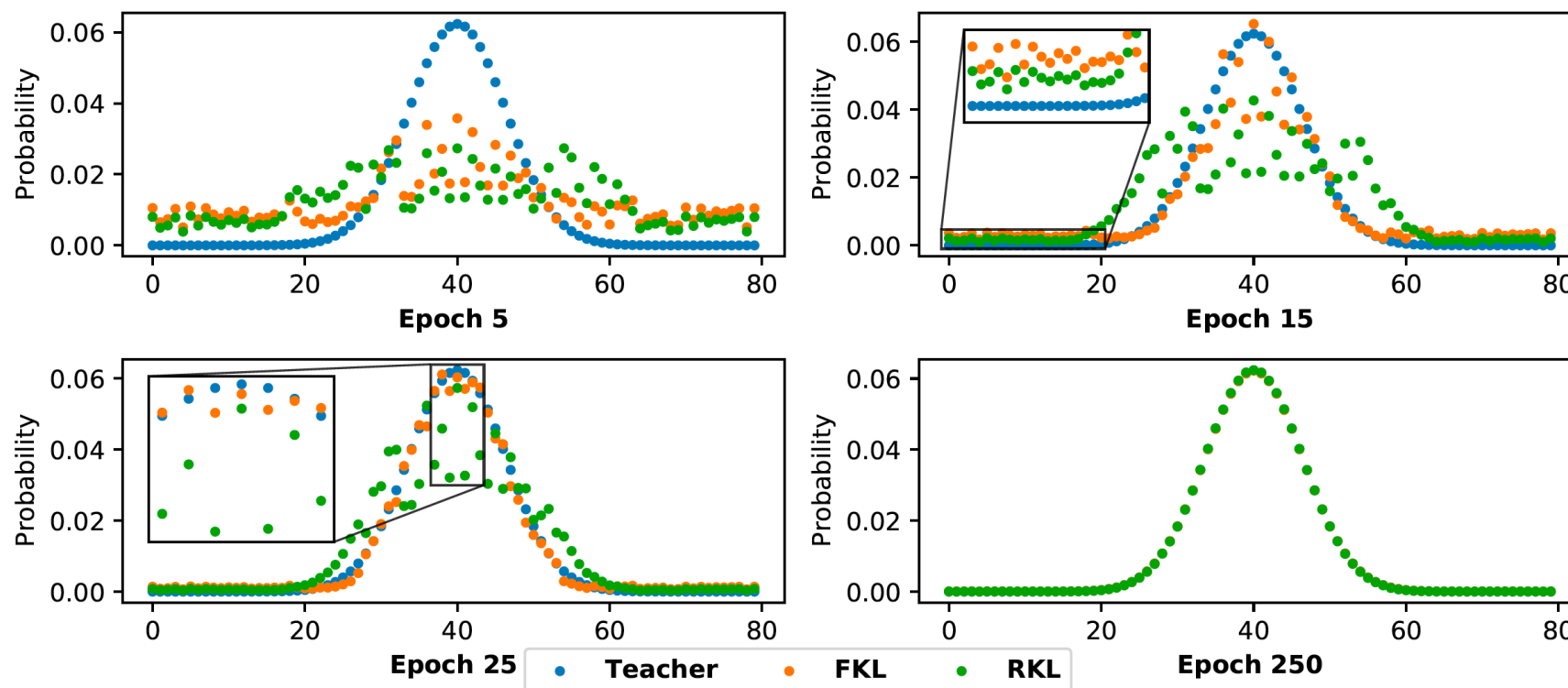


Figure 3: The distributions at various epochs for FKL and RKL on toy data, where the teacher distribution and initial student distribution are the same. We can find that FKL focuses on the head part and RKL on the tail part at the beginning epochs, and both converge finally.

Adaptive Kullback-Leiber (AKL) divergence

Idea: assign larger weights to FKL if the gap on the head part is larger than the tail part, and vice versa

Indicate mask:

The gap g_{head} on the head part and g_{tail} on the tail part:

$$M[i] = \begin{cases} 1 & \text{when } Y_i \text{ belongs to head part} \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

$$\min \sum_{i=0}^V M[i] \quad s.t. \quad \sum_{j=0}^V M[j] p(Y_j | \mathbf{y}_{<t}) \geq \mu, \quad (12)$$

$$g_{head} = \sum_{i=0}^V M[i] \epsilon(p(Y_i | \mathbf{y}_{<t}), q_{\theta}(Y_i | \mathbf{y}_{<t})); \quad (13)$$

$$g_{tail} = \sum_{i=0}^V (1 - M[i]) \epsilon(p(Y_i | \mathbf{y}_{<t}), q_{\theta}(Y_i | \mathbf{y}_{<t})) \quad (14)$$

$\epsilon(\cdot)$ is the function to define the point-wise gap.

Then AKL is defined as

$$AKL(p, q_{\theta}) = \frac{g_{head}}{g_{head} + g_{tail}} FKL(p, q_{\theta}) + \frac{g_{tail}}{g_{head} + g_{tail}} RKL(p, q_{\theta}). \quad (15)$$

μ as 0.5 and the gap function $\epsilon(p(z), q(z)) = |p(z) - q(z)|$

Experiments

Method	GPT2 1.5B → GPT2 120M			LLaMA 6.7B → TinyLLaMA 1.1B		
	Dolly (500)	S-NI (1694)	UnNI (10000)	Dolly (500)	S-NI (1694)	UnNI (10000)
Teacher	26.98±0.27	27.25±0.16	31.61±0.16	26.73±0.65	32.75±0.24	34.61±0.08
SFT	23.01±0.18	16.48±0.28	18.43±0.09	22.05±0.38	27.79±0.20	25.96±0.13
SeqKD	23.30±0.38	16.35±0.17	18.51±0.04	22.67±0.57	26.97±0.32	27.35±0.06
FKL	23.46±0.56	16.63±0.48	19.27±0.06	22.24±0.38	28.07±0.41	26.93±0.09
RKL	22.62±0.34	17.88±0.17	19.35±0.06	23.95±0.53	28.90±0.41	27.89±0.08
FKL+RKL	23.36±0.53	17.83±0.23	20.37±0.08	24.08±0.51	30.98±0.31	30.48±0.10
AKL (Ours)	23.88±0.46	19.15±0.21	21.97±0.13	24.40±0.42	31.37±0.23	31.05±0.17

Table 1: The results of Rouge-L scores under different distillation settings. Our proposed AKL outperforms all baselines without any extra parameters, especially in the larger UnNI dataset.

SFT: directly train the student model on the dataset without distillation

SeqKD (Kim and Rush, 2016): train the student model on the data generated from the teacher model

FKL+RKL: simply combine forward KL and reverse KL via $0.5\text{FKL} + 0.5\text{RKL}$

Rouge-L score: <https://zhuanlan.zhihu.com/p/543103440>

Experiments

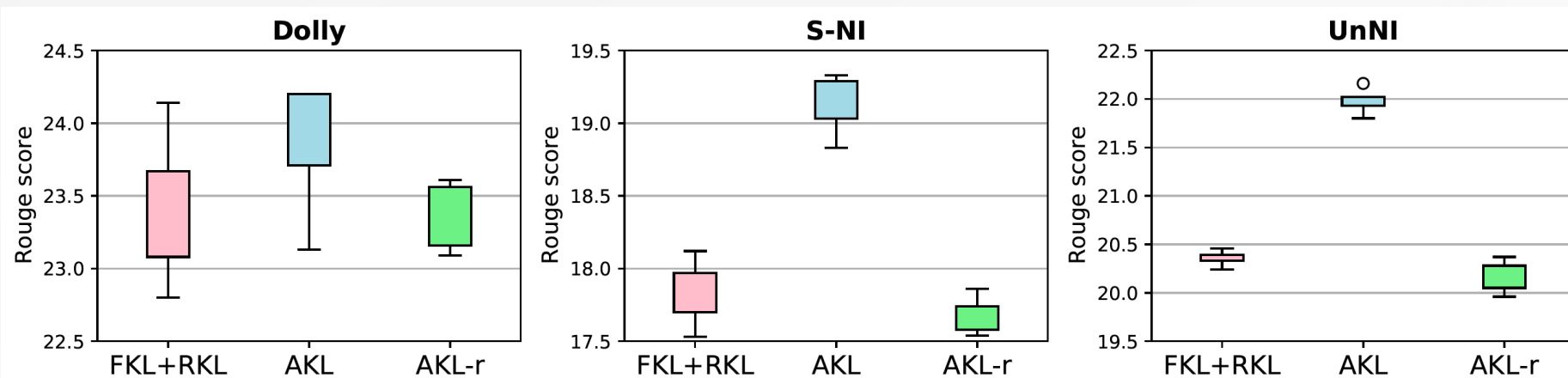


Figure 4: The results of FKL+RKL, proposed AKL, and AKL-r on GPT 2 120M. After flipping the loss weight, AKL-r performs worse than AKL on all three datasets.

Ablation: Flipping the Loss Weights

$$AKL-r(p, q_{\theta}) = \frac{g_{tail}}{g_{head} + g_{tail}} FKL(p, q_{\theta}) + \frac{g_{head}}{g_{head} + g_{tail}} RKL(p, q_{\theta}). \quad (16)$$

Experiments

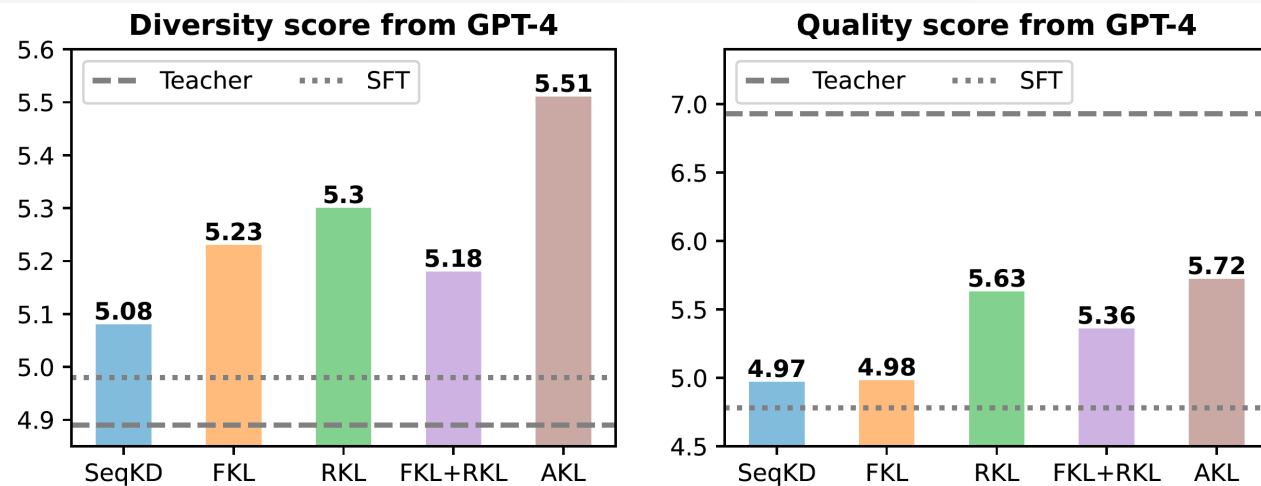


Figure 5: The scores of diversity and quality from GPT-4 (total score: 10) on the responses from TinyLLaMA. AKL can improve the diversity and quality of generated responses compared to the baselines.

Evaluation via GPT-4

Setting	Win/Tie/Loss Ratio	
	Diversity	Quality
AKL vs Teacher	36.1/44.4/19.5	15.9/35.6/48.5
AKL vs SFT	30.6/54.2/15.2	46.7/43.0/10.3
AKL vs SeqKD	33.3/48.6/18.1	39.3/43.0/17.7
AKL vs FKL	25.0/58.3/16.7	39.2/47.7/13.1
AKL vs RKL	30.6/47.2/22.2	27.1/48.6/24.3
AKL vs FKL+RKL	31.9/55.6/12.5	27.1/57.0/15.9

Table 3: Win/Tie/Loss ratios comparing AKL against baselines.

User Study



THANKS!

